Project Overview

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TEAM B1

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1. Introduction

This document provides a general overview of the project delivered to CERcuits. It describes the context in The project was developed, the objectives our team aimed for, the scope of the work, and the outcomes we achieved.

2. Project context

2.1. The Problem

CERcuits identified major bottlenecks in its handling incoming email communication and support queries. In their current state they have a single technical sales engineer who is responsible for processing the full volume of support emails manually.

Handling emails manually is time-consuming, prone to errors, and difficult to scale as the volume of communication increases. Important can be missed, responses become more inconsistent as the under higher workload, and response times can vary depending on the employee's workload.

CERcuits also wanted a solution that can support employees without fully automating decision-making, maintaining control, traceability and reliability.

2.2. Initial goals

Our initial goals of the project were to:

- Assist engineers by generating draft responses to customer emails and support questions.
- Use the company's documents as the primary knowledge base through RAG (Retrieval-Augmented Generation).
- Allow full control and management over said system by an admin.
- Create a maintainable and extendable technical proof of concept.
- Deliver clear documentation to allow future maintenance and development by CERcuits.

3. Scope of Project

3.1. What was included

The project includes:

- An AI backend system for processing user queries and email content.
- Context-aware response generation grounded in the company provided documents.
- Ability to view analytics about the usage of the application.
- Access control via managing user accounts.
- Validation and security mechanisms to prevent prompt injection and misuse.

3.2. What was excluded

The following aspects were explicitly outside the existing scope of this project:

- Full automation of email sending (responses are drafted, not sent automatically).
- Training of custom AI models beyond configuration, orchestration, and RAG.

3.3. Assumptions

The project was developed under the following assumptions:

- The system will be operated by trained users within CERcuits with a solid amount of technical knowledge.
- The required infrastructure to run the application and its related AI components is avaliable.
- AI-generated output will be reviewed by users before it is used.

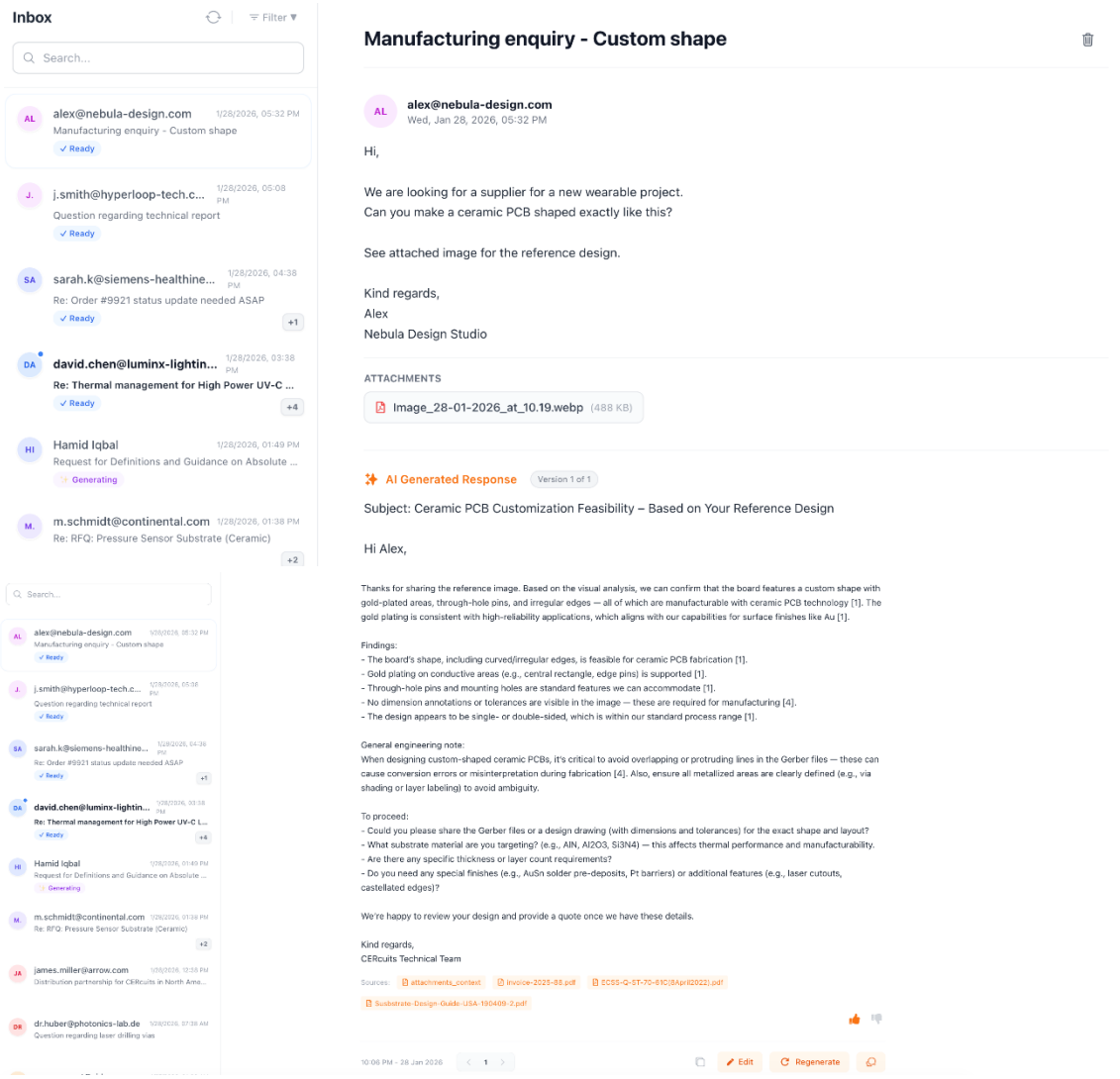
4. What we Accomplished

The primary deliverable of this project is a fully functional AI-assisted application designed to support and improve the handling of email communication and user queries within CERcuits.

4.1. Main features

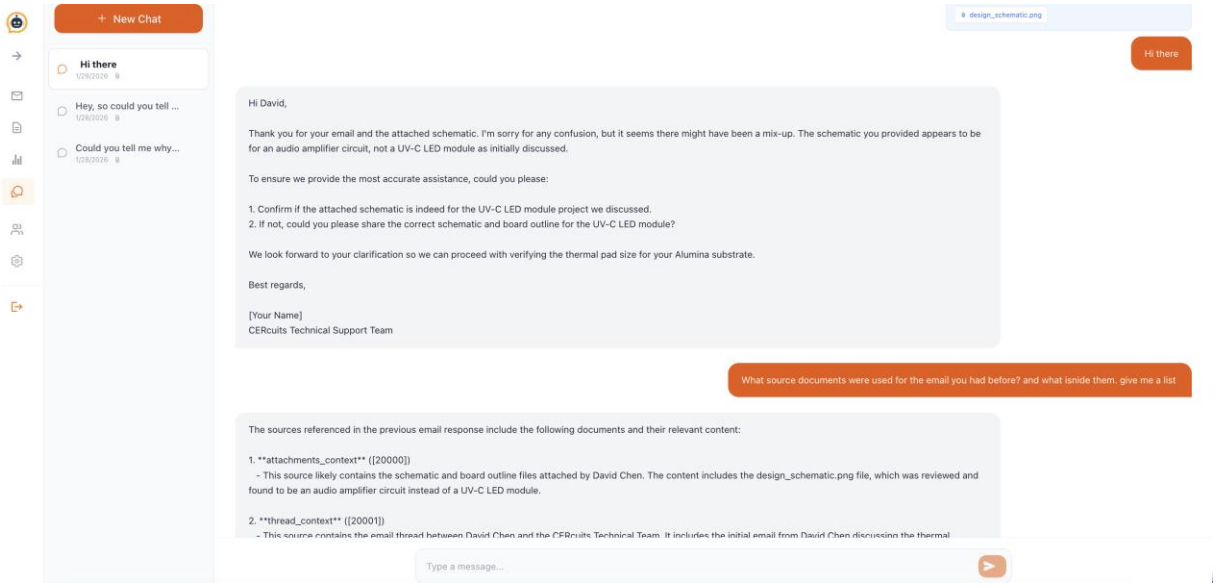
4.1.1. Context-Aware email generation

The system can generate draft replies to customer emails using RAG. It retrieves the most relevant information from the uploaded knowledge base and uses it as grounding for the response. For emails and documents that include images, technical drawings, diagrams, tables, etc., the vision component extracts meaningful content, so no context is overlooked. This helps engineers reply faster and more consistently while keeping the user in control.



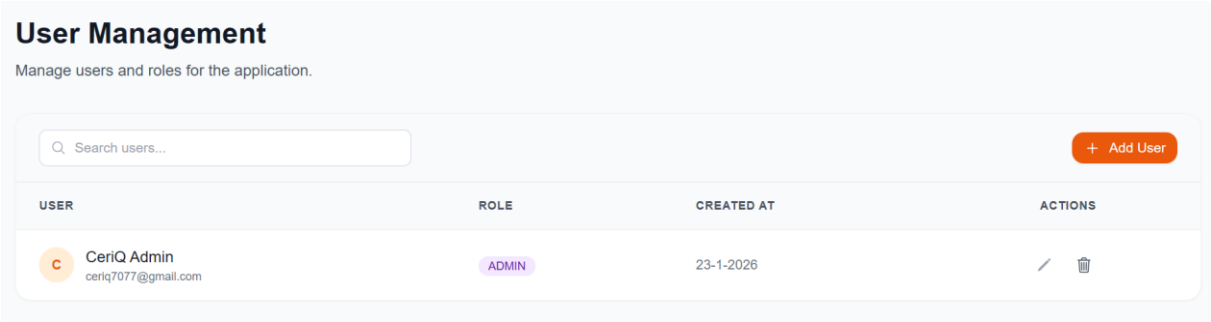
4.1.2. CERcuits AI chatbot

The chatbot allows users to have a natural conversation with the assistant either around a specific email or independently. In an email-driven flow, you can select an email and then ask follow-up questions such as why the assistant used specific sources, which documents influenced the draft or what information was missing. You can also use the same chat interface without selecting an email to general questions and get structured answers in the same support-engineer style. This function adds transparency and makes it easier to review and trust the generated response.



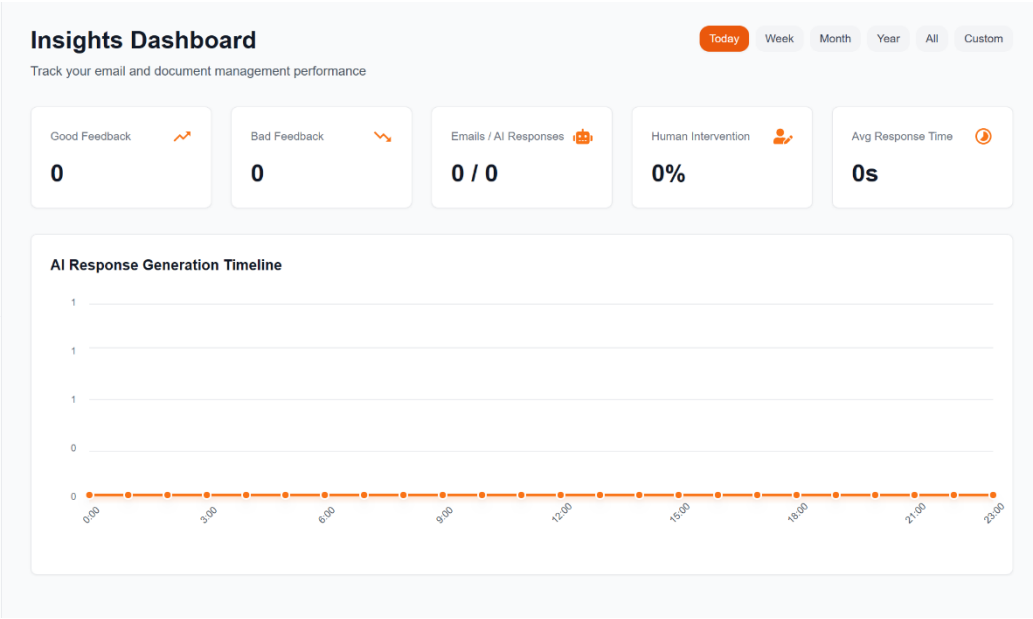
4.1.3. User management

Our user management system facilitates the creation and management of new user accounts by an admin user. The addition of the functionality allows for stricter access control, operational management, and analytical logging.



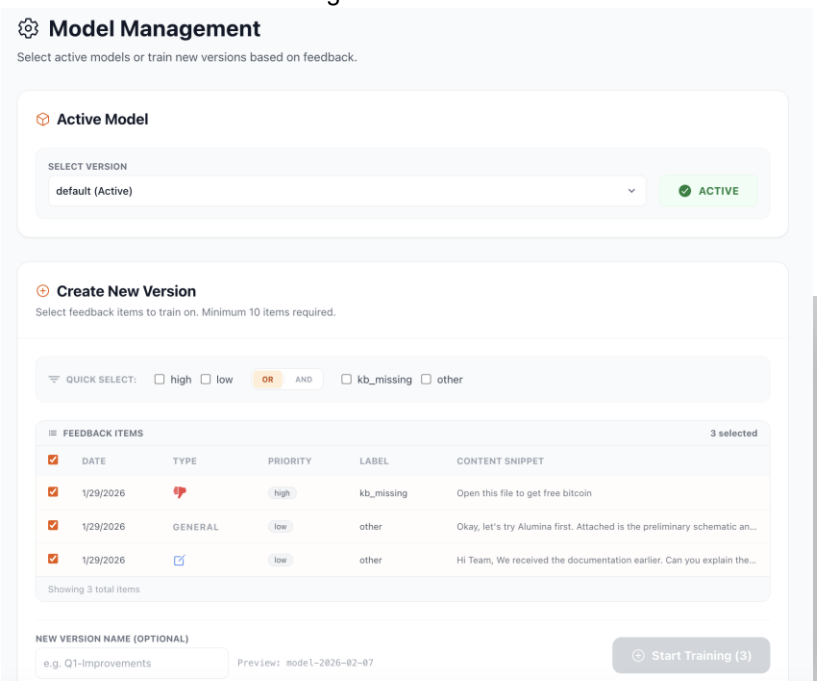
4.1.4. AI insights

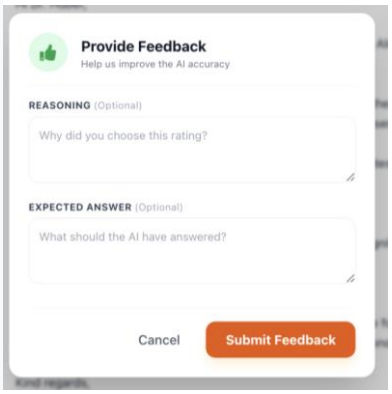
As requested by the client, a short page showing insights about the AI's usage was implemented. This page shows you when the AI has been used, what the users of the system thought about the AI's performance, and some minor insights about response times.



4.1.5. Continues improvement via feedback

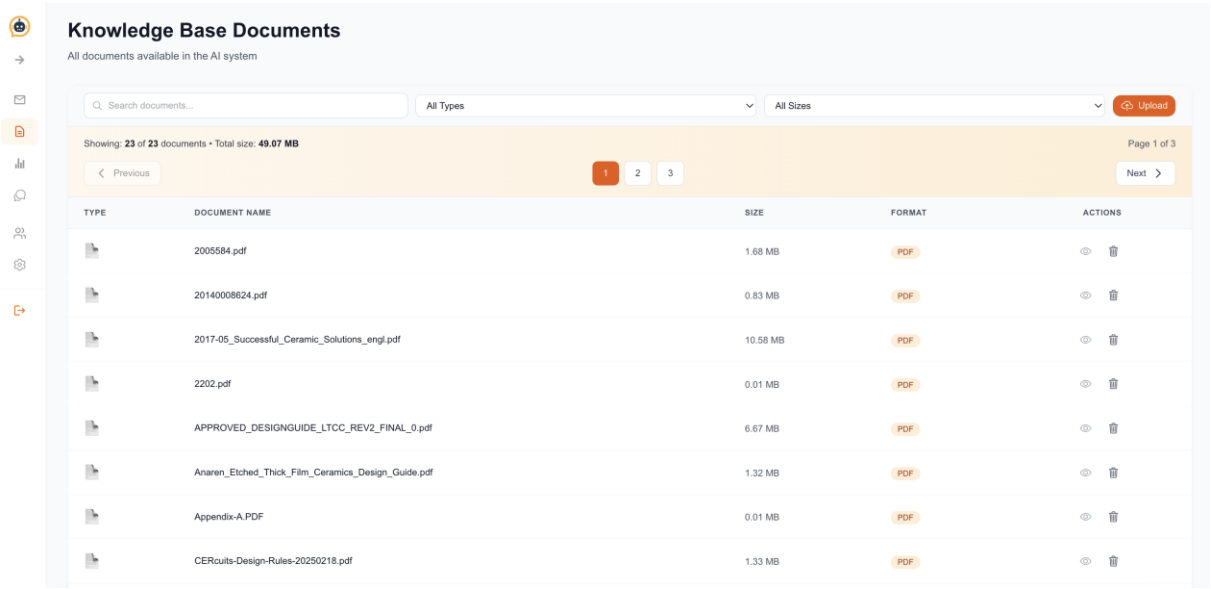
Users using the system can give feedback back to the AI system based upon a response it generated to an email. This feedback can later be used by an admin to improve the AI system by creating new "models" using that feedback which the AI will use in generating its future responses.
It's also noteworthy that the admin is able select old models, delete existing ones, and select limited feedback elements for training based on a classified label.





4.1.6. Knowledge base

Knowledge base is a document library that powers our RAG workflow. The files can be managed, so when a new file is added, the system automatically runs an ingestion pipeline to store the resulting vectors in the vector database, ready to be referenced. When a file is deleted, its corresponding vectors are also removed from the vector database. This setup keeps the assistant's knowledge traceable, editable, and most importantly: always up to date.

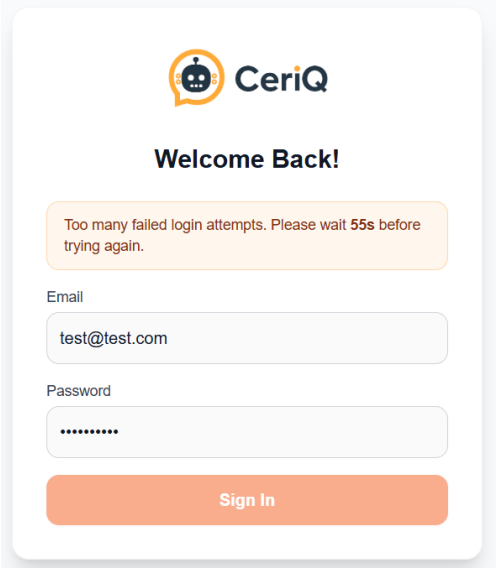


4.1.7. Application security

The security is designed to protect user identity and system integrity. All user interactions, including login, are encrypted via HTTPS to ensure credentials are never intercepted. Passwords are immediately hashed and salted using the Bcrypt algorithm before being stored in the database, ensuring that raw passwords never exist. Once authenticated, user sessions are managed using secure access tokens, eliminating the need for vulnerable server-side session storage.

We enforce strict Role-Based Access Control (RBAC) to maintain privilege separation. Standard users are restricted from accessing administrative functions such as the User Management Panel and Model Management. These sensitive areas are locked down to administrator accounts only.

To prevent abuse and ensure availability, the system actively monitors incoming traffic. We implement Rate Limiting to stop brute-force attacks and Denial-of-Service (DoS) attempts. Additionally, the application employs proactive IP Blocking, automatically banning suspicious IP addresses and enforcing geo-restrictions to block traffic from non-European regions, significantly reducing the external attack surface.



4.2. Technical milestones

4.2.1. AI pipeline architecture

We made clear architectural decisions to support maintainability and future expansion of the AI system. This was achieved by using a pipeline-centric architecture and applying loose coupling and dependency injection principles where applicable.

With this approach, it was simple to implement extra steps or preprocessing, validation, security, and more across all pipelines. It also makes it easier to introduce new pipelines or extend existing ones without rewriting large parts of the codebase.

4.2.2. Security implementations

To realize this strategy, we deploy a robust stack of active defence technologies. At the network perimeter, an Nginx reverse proxy handles encryption and enforces rate limits. Behind this, Fail2Ban and CrowdSec work in tandem to actively monitor traffic. These tools enforce our geo-blocking policy and block known malicious actors based on global reputation lists. Internally, content safety is managed via a privacy-preserving hybrid workflow: incoming email attachments are scanned locally by ClamAV for immediate threat detection. Simultaneously, a cryptographic hash of the file, never the file itself is verified against VirusTotal, allowing the system to leverage external threat intelligence without exposing confidential documents. Finally, data resilience is guaranteed through automated daily backups with a strict 14-day retention policy.

The system is engineered for strict regulatory compliance and data privacy. Unlike cloud-based solutions, all core data processing including AI inference, vector embedding, and database storage occurs entirely within our private infrastructure. No customer data or internal documents are ever sent to external providers, ensuring complete data isolation. By strictly geo-blocking non-European traffic and hosting all services on-premises, the deployment aligns with GDPR requirements regarding data residency and cross-border transfer restrictions.

Security begins before the application is even deployed. We utilize an automated CI/CD pipeline that enforces security standards on every code update. Static Application Security Testing (SAST) tools like SonarQube and Semgrep analyse source code for vulnerabilities before they can be merged. Additionally, we use Trivy to scan our Docker container images for known system vulnerabilities (CVEs) and enforce automated Secret Detection to prevent the accidental exposure of API keys or credentials in the codebase.

4.2.3. Documentaion and handover readiness

All major technical decisions, configurations, and workflows were documented. This milestone ensures that our system can be fully understood and maintained by developers who were not involved in the original implementation.

5. Current Project Status

5.1. Development state

At the time of delivery, the project is in a stable and functional state as proof of concept. All core features and more have been implemented, tested, and documented. The system is suitable for controlled internal use within CERcuits and provides a solid foundation for future development or scaling.

5.2. Unexpected changes

In our analysis we layout the tools we intended to use for this project, what features and intended to implement, And what we deemed strictly “out of scope”. We managed to stick to many of our original design decisions, but there are some key changes that we had not originally planned before.

5.2.1. Addition of the AI chatbot

During out analysis we had strictly deemed the created of a chatbot as “Out of scope”. We intended to put all our efforts into having the chatbot generate an optimal response to any email its giving instantly.

And yet, due to how well our AI system was progressing, in combination with feedback from our client. We decided to implement a chatbot after all. And we feel as thought we did as much with great success!

5.2.2. Switch from Lora feedback training to RAG

Originally, we intended to use a technique called “Lora” to continuously improve our AI system based on feedback. But after trying to apply a Lora solution as we had imagine in a local environment with all the architectural “pieces” of what we had imagined working together, we quickly ran into many issues.

This caused us to reevaluate how effective Lora is compared to simpler and conventional means of retraining like retraining via RAG. In doing so we found that for this use case, applying feedback based on a RAG system was actually drastically more effective than Lora.

This caused us to permanently pivot ourselves to a more RAG based solution to feedback then we had originally planned for. On reflection we feel as though this was a great decision. Since while being more effective it also made many other features like model switching drastically simpler to implement.

5.3. Known limitations

While the system is fully functional, a few limitations remain. The quality of the AI-generated responses depends heavily on the accuracy and completeness of the contextual documentation provided in the knowledgebase. The assistant can reference its general knowledge, but it heavily prefers a complete answer grounded in its context.

In addition, there are several smaller limitations of the current AI-system:

- Our Chatbot does not have access to the knowledge base beyond sources mentioned in its context. (e.g., used in an email is referencing)
- Our Chatbot does not have access to attachments of emails in its context.
- Feedback for the AI system is not collected in any way beyond the feedback buttons.
- The system does not have a “global context”. Beyond email chains the system does not remember context.

Additionally, the system has not been tested under a simulated real life workload. As a result, a full scale deployment of beyond a small team in CERcuits will require further development.

For these reasons human review of the AI output is strongly recommended.

6. Summary

This project delivers a secure and extensible AI-assistant solution tailored to operational needs and requirements laid out by CERcuits. The primary goal was to reduce the workload associated with manual email handling while maintaining human control, reliability, and transparency. The final system successfully supports engineers by generating context-aware email drafts grounded in CERcuits’ internal documentation using a retrieval-augmented generation (RAG) system.

In addition to the email assistant, the CERcuits AI chatbot provides a conversational interface that improves transparency and usability. This increases trust in the AI output and makes the system easier to integrate into daily workflows.

From a technical perspective, the solution is built in a modular and maintainable architecture that supports future expansion. Strong emphasis was placed on security and data privacy, with role-based access control, encrypted communication, and active threat modeling.

At the time of delivery, we feel that the application is stable and fully functional as a proof of concept for internal use withing CERcuits. While some limitations remain, the system provides a solid foundation for further development.